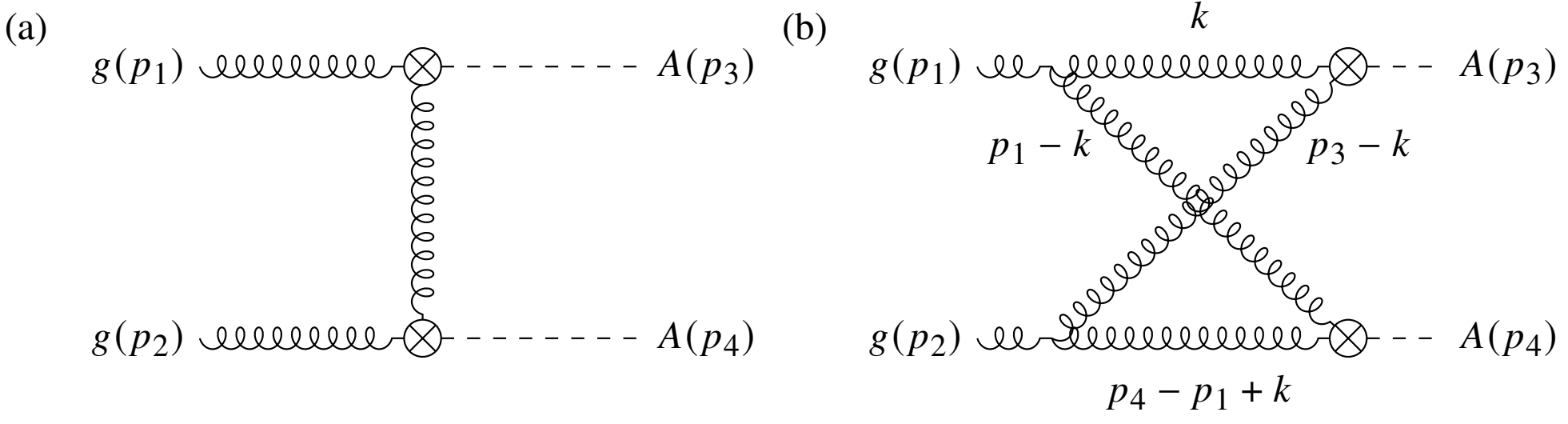
(a)
$g(p_1)$
$A(p_3)$
$g(p_2)$
$A(p_4)$
(b)
k
$g(p_1)$
$A(p_3)$
$p_1 - k$
$p_3 - k$
$g(p_2)$
$A(p_4)$
$p_4 - p_1 + k$